MUNICIPAL WASTE COLLECTION OPTIMISER

NAME : Y.Sahasra

ROLL NUMBER : 2620030143

DEPARTMENT : CSE

SUBJECT : PSPJ

Abstract

The Municipal Waste Collection Optimiser is a software system designed to improve the efficiency of municipal waste collection in urban areas. In traditional waste collection, vehicles often follow fixed routes even when some bins are half-empty while others are overflowing. This project analyses the waste level at different collection points and gives priority to locations with higher waste levels. It helps organize the collection order, reduce unnecessary trips, save fuel and time, and minimize waste overflow. The system can also classify collection points based on waste levels and provide an optimized collection plan.

Advantages

- Reduces Waste Overflow – High-level bins can be collected first, reducing overflowing waste.
- Saves Fuel – Better collection planning can reduce unnecessary vehicle travel.
- Saves Collection Time – The system provides a priority-based collection order.
- Efficient Resource Utilization – Collection vehicles can be used more effectively.
- Reduces Operational Cost – Efficient routes and fewer unnecessary trips can reduce collection expenses.
- Handles Multiple Collection Points – The system can manage waste information from different locations and zones.
- Priority-Based Collection – Locations can be classified as High, Medium, or Low priority.
- Supports Cleaner Cities – Timely collection helps maintain cleaner streets and public areas.

Implementation

The Municipal Waste Collection Optimiser can be implemented using Java.

1. Input Collection Data

The system accepts collection point/location, waste quantity or bin fill level, waste type, and collection priority.

2. Store the Data

Arrays or other data structures can be used to store information about multiple collection points.

3. Calculate Priority

The system checks the waste level and assigns priority.

- Waste Level >= 80% → HIGH
- Waste Level 50–79% → MEDIUM
- Waste Level < 50% → LOW

4. Optimise Collection Order

Collection points are arranged according to their priority and waste level so that locations requiring urgent collection can be visited first.

5. Track Collection

After a bin is collected, its waste level can be updated to indicate that the collection has been completed.

6. Generate Report

- Total waste collected
- Number of locations collected
- High-priority locations
- Missed or overflowing bins
- Collection order
- Route efficiency

Technologies Used

- Language: Java
- Input: Scanner
- Storage: Arrays / Data Structures
- Decision Making: if-else
- Repetition: for / while loops
- Sorting: Comparison and sorting logic

- Output: Console Report

- ## Basic Implementation Flow

Enter Collection Points

↓

Enter Waste Levels

↓

Check Bin/Waste Status

↓

Assign Priority

↓

Arrange Collection Order

↓

Collect Waste

↓

Update Waste Levels

↓

Generate Collection Report

Project Comic / Storyboard

Panel 1 – The Problem

A municipal worker finds one nearly empty bin and another overflowing bin while the collection vehicle follows a fixed route.

Panel 2 – Waste Data

The system records the collection locations and checks the waste level of each bin.

Panel 3 – Priority

The optimiser identifies the overflowing bin as HIGH priority and arranges the collection order.

Panel 4 – Collection

The collection vehicle follows the planned order and collects the high-priority waste first.

Panel 5 – Result

Waste overflow is reduced and the collection process becomes more organised and efficient.

OUTPUT :

```
' 'MunicipalWasteCollectionOptimiser'

==================================
       MUNICIPAL WASTE AUTHORITY
==================================
1) Update fills
2) Build route
3) Collect
4) Report
5) Exit
Choice: 2

Truck capacity: 1000.0 L
Collection threshold: 70.0%

Priority bins:
B02(75.0%)
B04(88.0%)

Route:
Depot -> B04 -> B02
Route distance: 8.4 km
Truck load: 740.0 L / 1000.0 L
```

Conclusion

The Municipal Waste Collection Optimiser provides an efficient way to manage waste collection from multiple locations. By analysing waste levels and prioritising areas with higher waste, the system helps reduce overflowing bins, unnecessary trips, fuel consumption, and collection time. It also improves the utilisation of collection vehicles and resources. Overall, the project demonstrates how a simple software-based approach can support cleaner cities, efficient waste management, and sustainable urban development.

Project Summary

- Identifies waste levels at different collection points.
- Assigns collection priority based on waste level.
- Organises the collection order.
- Helps reduce overflowing waste.
- Supports efficient use of collection vehicles and resources.
- Provides a simple Java-based approach to municipal waste management.